EcoQuest — Viva Voce Preparation & Project Defense Guide 🎓

B.Sc. Computer Science — Community Engagement Project (CEP)
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Theme: Gamified App for Sustainability
Platform: React Native (Expo SDK 57) for Android

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Section A: One-Minute Project Introduction

Speak this naturally and confidently when the examiner says: "Aditya, introduce your project."

"Good morning, Respected Examiners. My name is Aditya Deonarayan Jha, and for my Community Engagement Project in Third Year B.Sc. Computer Science, I have developed **EcoQuest — Campus Restoration Game**.

On college campuses, sustainability initiatives like switching off empty classroom fans, segregating waste, saving water, and avoiding single-use plastics are usually communicated through static notice boards and circulars. Students know these habits are important, but without immediate feedback or visible results, engagement remains low.

EcoQuest solves this through **gamification**. The app turns campus sustainability into an interactive restoration project. When a student completes an everyday eco-challenge, they earn **Lifetime Eco Points** and build a **Spendable Balance**. They reinvest their spendable balance into physical campus projects—like planting shade trees, installing rooftop solar panels, deploying compost bins, and building water refill stations. As projects are funded, the campus illustration visibly transforms on screen from a dormant quad into a flourishing green sanctuary.

The app is built with **React Native and Expo SDK 57**, uses **React Context** for unified state management, and persists data offline using **AsyncStorage**. It features eight quests, four restoration upgrades, four player levels, eight badges, and a dynamic leaderboard.

As a third-year demonstration prototype, EcoQuest focuses on incentive modeling and user experience. Quests are self-reported and peer scores are simulated, designed to prove how gamified software can motivate sustainable habits without requiring expensive cloud infrastructure."

Section B: Three-Minute Technical Explanation

Use this structured response if the examiner asks: "Explain the technical architecture and how data flows through your app."

"EcoQuest is developed as a client-rendered, offline-first mobile application using **React Native 0.86** and **Expo SDK 57**, targeting Android devices.

1. Navigation & UI Structure
The user interface is organized into a clean **five-tab navigation layout**:
Home Tab: *Features our custom* ***CampusScene*** *illustration, real-time level progress, glanceable points, and the sequential campus restoration upgrade card.*
Quests Tab: Displays eight predefined sustainability challenges across five environmental categories—Energy, Waste, Water, Transport, and Food—with category filtering and one-tap completion.
Rank Tab: *Features an eight-participant dynamic campus leaderboard and a visual Top-3 podium.*
Badges Tab: Displays eight achievement trophies with real-time unlock tracking.
Profile Tab: *Shows student statistics and a two-step confirmed Demo Reset button.*

2. State Management & Single Source of Truth
All global state is managed through React Context in `GameContext.js` *and consumed across screens via our custom* `useGame()` *hook.*

To eliminate data anomalies, we strictly adhere to the ***Single Source of Truth*** *principle. Rather than storing computed values like current score, stage number, badge unlocks, or leaderboard ranks, we only store two minimal arrays of primary ground-truth records:*
1. `completedQuestIds` *: Array of unique quest ID strings.*
2. `purchasedUpgradeIds` *: Array of unique sequential upgrade ID strings.*

All other values are dynamically derived through ***pure selector functions****:*
Lifetime Points: Calculated as the sum of rewards from completed quests. This number never decreases.
Total Upgrade Cost: *Calculated as the sum of costs for purchased campus upgrades.*
Spendable Points: Calculated as `Math.max(0, lifetimePoints - totalUpgradeCost)` .
Restoration Stage: *Directly equals the count of purchased upgrades (0 to 4).*
Player Level: Derived from lifetime points across four tiers (Eco Seedling, Green Scout, Earth Guardian, Planet Champion).
Badges & Leaderboard: *Evaluated dynamically from lifetime points and quest completion history.*

3. Complete Information Flow
When a user taps 'Complete Quest' in the Quests tab:
1. `completeQuest(questId)` *verifies that the quest exists and has not already been completed.*
2. *The quest ID is appended to* `completedQuestIds` *in both React state and an atomic synchronous* `useRef` .
3. *State re-renders:* `lifetimePoints` *and* `spendablePoints` *update instantly.*
4. *If spendable points now meet the cost of the next sequential upgrade, the upgrade button on the Home tab lights up.*
5. *When the user taps 'Fund Upgrade',* `purchaseNextUpgrade()` *validates that the user can afford it, appends the upgrade ID to* `purchasedUpgradeIds` *, and increments* `restorationStage` .
6. `CampusScene.js` *receives the updated stage (for example, Stage 1) and conditionally renders native visual elements—changing the quad ground from dry beige to lush green and adding native oak trees.*

4. Offline Persistence & Hydration
Data is saved locally using `@react-native-async-storage/async-storage` under key `'ecoquest.gameState.v1'`.
During startup, an `ActivityIndicator` loading splash card guards the UI until `hydrate()` reads and sanitizes stored JSON.
Writes are serialized through a Promise queue (`saveChainRef`), preventing race conditions during rapid taps.
If the presenter taps 'Reset Everything' in the Profile tab, a native confirmation modal clears the storage key and resets both arrays to empty, cleanly returning the app to Stage 0 for the next presentation."

Section C: Basic Viva Questions & Answers

Q1: What is EcoQuest?

Answer: EcoQuest is a gamified sustainability mobile application designed for college campuses. It encourages students to adopt eco-friendly habits by rewarding completed micro-quests with points that restore a virtual campus environment.

Q2: Why was this project selected for your Community Engagement Project (CEP)?

Answer: Campus environmental initiatives often fail due to passive communication. We selected this project to bridge the gap between environmental awareness and daily student action using interactive computer science principles.

Q3: What core problem does EcoQuest address?

Answer: It addresses the "drop in the ocean" perception—where students feel their individual green actions (like turning off a fan or carrying a bottle) make no visible difference. EcoQuest makes individual effort measurable and visually rewarding.

Q4: What is gamification in software engineering?

Answer: Gamification is the integration of game-design elements—such as points, badges, levels, and progress visualization—into non-game contexts to boost user engagement, motivation, and habit formation.

Q5: Why did you choose React Native?

Answer: React Native allows building cross-platform native mobile applications for Android and iOS using JavaScript and React. It provides native performance, reusable component architecture, and rapid UI development.

Q6: What is Expo and why is it used?

Answer: Expo is an open-source framework and platform built around React Native. It provides a managed workflow, pre-configured build tools, and the Expo Go client app, enabling quick testing and live deployment directly on physical mobile phones.

Q7: What is React Context?

Answer: React Context is a built-in React feature that allows data to be shared across the entire component tree without having to manually pass props down through every level (prop drilling).

Q8: What is a React component?

Answer: A component is an independent, reusable piece of user interface code. In React, components are written as JavaScript functions that take inputs (props) and return JSX describing what should appear on the screen.

Q9: Why was JavaScript chosen over TypeScript?

Answer: JavaScript is the native language of the web and React Native. For a semester-length demonstration project, standard ES6+ JavaScript minimized compilation overhead and allowed rapid iteration while maintaining complete architectural cleanliness.

Q10: What is AsyncStorage?

Answer: AsyncStorage is an unencrypted, asynchronous, persistent key-value storage system for React Native. It stores data locally on the mobile device, allowing game progress to persist across app restarts without an internet connection.

Q11: What is the purpose of `src/context/GameContext.js`?

Answer: It is the central nervous system of the application. It holds global state, provides calculation selector functions, manages local persistence via AsyncStorage, and exposes the `useGame()` hook to all screens.

Q12: What is the purpose of `src/data/gameData.js`?

Answer: It is the single static source of truth for all game rules and configurations: the 8 sustainability quests, 4 campus upgrades, 4 player levels, 8 achievement badges, and the initial campus leaderboard roster.

Q13: Why is `CampusScene.js` separated from `HomeScreen.js`?

Answer: To maintain the **Separation of Concerns** principle. `CampusScene.js` is a dedicated presentation component responsible strictly for drawing the illustrated campus environment based on the `restorationStage` prop, keeping `HomeScreen.js` focused on layout and user controls.

Q14: What happens under the hood when a user completes a quest?

Answer: The `completeQuest(questId)` function checks whether the quest was already completed. If not, it appends the quest ID to `completedQuestIds`, recalculates lifetime and spendable points, updates the UI, and persists the new ID array to AsyncStorage.

Q15: Why can a quest be completed only once in this demo?

Answer: In this demonstration prototype, each quest offers a one-time reward so examiners can easily verify economic balance (210 total points available) and test deterministic progression without infinite loop inflation.

Q16: How does a campus upgrade work?

Answer: Upgrades must be unlocked sequentially. When the player has accumulated enough spendable points to cover the next upgrade's cost, the 'Fund Upgrade' button activates. Tapping it deducts the cost from spendable points, increments the restoration stage, and visually transforms the campus quad.

Q17: What is the difference between Lifetime Points and Spendable Points?

Answer:

- **Lifetime Points:** The grand cumulative total of all points ever earned from quests. It **never decreases**, and it drives player levels, badges, and leaderboard rankings.
- **Spendable Points:** The current available currency. It increases when quests are completed and decreases when campus upgrades are funded.

Q18: How are player levels determined?

Answer: Levels are derived strictly from Lifetime Points across four tiers: Level 1 (0–49 pts), Level 2 (50–99 pts), Level 3 (100–169 pts), and Level 4 (170–210 pts).

Q19: How are achievement badges unlocked?

Answer: Badges are automatically evaluated in `calculateBadges()` based on either score thresholds (100, 160, 200 pts) or specific quest category completions (e.g., completing 2 Waste quests unlocks Zero-Waste Pioneer).

Q20: How does the campus leaderboard work?

Answer: The leaderboard compares the demonstrator's lifetime points against seven benchmark student peers. As the user earns points, the list dynamically sorts in descending order, climbing from Rank #8 up to Rank #4.

Q21: What happens when the app restarts?

Answer: On startup, `GameContext` reads the serialized JSON from AsyncStorage under `'ecoquest.gameState.v1'`, sanitizes the IDs, recalculates all derived values, and renders the exact campus stage and points the user had previously earned.

Q22: Why is a reset confirmation dialog necessary?

Answer: Demo resets permanently erase all saved progress. To prevent accidental data loss during a live examination or demonstration, a two-step native alert (`Alert.alert`) requires the presenter to explicitly confirm before wiping storage.

Q23: How were the game calculations and economy tested?

Answer: We wrote an automated integration test suite in `tests/test_integration.mjs` containing 83 assertions. It executes via Node.js and verifies quest point totals, upgrade costs, level transitions, badge triggers, serialization, and reset behaviors with 100% pass rate.

Q24: What are the primary limitations of the current app?

Answer:

1. Quests are self-reported via button clicks without hardware verification.
2. Leaderboard peers are simulated benchmarks rather than live network players.
3. Storage is local to the device without cloud sync.
4. Environmental impact metrics are pedagogical estimates.

Q25: What would be a sensible future improvement?

Answer: Integrating camera-based image classification using an AI model (like TensorFlow Lite) so students can take a photo of their reusable bottle or segregated bin to automatically verify quest completion.

Q26: What is the difference between React Native and standard React for the web?

Answer: React web renders HTML elements (, , ,) to the browser DOM. React Native renders native mobile widgets (, ,) using platform-specific native UI components on Android and iOS.

Q27: How does EcoQuest handle safe areas on Android devices?

Answer: We use `react-native-safe-area-context`. The root `App.js` wraps the tree in `SafeAreaProvider`, and `BottomNavBar.js` dynamically queries `useSafeAreaInsets()` to add safety padding over Android 3-button navigation bars and gesture navigation pills.

Q28: What is Expo Go?

Answer: Expo Go is a mobile client application provided by Expo that allows developers to run and test React Native projects on a physical phone immediately over Wi-Fi without compiling native Android APK binaries locally.

Section D: Technical Architecture Questions

> **Q1: Describe the architectural pattern used in EcoQuest.**

Answer: EcoQuest follows a **Unidirectional Data Flow (Flux-like)** architecture within React Native. A central provider (`GameProvider`) acts as the single source of truth. State updates trigger pure derivation functions (selectors), which pass clean computed props down to consumer screens. Actions flow up through explicitly exposed handler functions (`completeQuest` , `purchaseNextUpgrade` , `resetDemo`).

> **Q2: What is the difference between shared context state and local component state in this project?**

Answer:

- **Shared Context State:** Holds data needed across multiple tabs—specifically `completedQuestIds` and `purchasedUpgradeIds` . This is managed globally in `GameContext.js` .
- **Local Component State:** Holds UI-only, ephemeral state relevant to a single component—such as `activeTab` in `AppContent` (determining which screen is visible) or `selectedCategory` in `QuestsScreen.js` (filtering displayed quest cards).

> **Q3: Explain how `createContext` and `useContext` are implemented.**

Answer: In `GameContext.js` , we create the context using `const GameContext = createContext(null)` . We wrap the application in `GameContext.Provider` passing the `contextValue` object. To consume the context safely, we export a custom hook `useGame()` which calls `useContext(GameContext)` and throws a descriptive error if called outside a `GameProvider` .

> **Q4: What is the "Primary Records versus Derived State" pattern, and why was it used?**

Answer: Storing computed values (like score, level, badge status, or stage) in database storage creates data duplication and synchronization bugs (e.g., score says 50, but quests only add up to 30). In EcoQuest, we **only persist the minimal ground-truth records**: `completedQuestIds` and `purchasedUpgradeIds` . Everything else is derived on the fly, guaranteeing 100% mathematical consistency.

> **Q5: What are pure calculation selector functions?**

Answer: A pure function is a function where the return value is determined solely by its input arguments, with zero side effects. Functions like `calculateLifetimePoints(ids)` , `calculateSpendablePoints(qIds, uIds)` , and `calculateLevelInfo(pts)` take inputs and return results deterministically, making them easy to unit test and verify.

> **Q6: How does the application protect against duplicate quest completion?**

Answer: Protection exists at three layers:

1. UI Layer: The button on the quest card is disabled and displays a green checkmark "Completed ✓".

2. Context Layer: `processQuestCompletion()` checks `completedQuestIds.includes(questId)` . If present, it rejects the operation with `{ success: false, reason: 'already_completed' }` .

3. Reference Layer: Synchronous `completedQuestIdsRef` checks prevent rapid double-tap race conditions before React re-renders.

> **Q7: What are the economic invariants enforced by the game engine?**

Answer:

1. `lifetimePoints >= spendablePoints` at all times.

2. `lifetimePoints` monotonically increases or stays constant; it **never decreases**.
3. `spendablePoints >= 0`; it can never be negative.
4. Total upgrade cost across all 4 stages is strictly 180 points.
5. All 8 quests yield exactly 210 points, leaving a guaranteed positive surplus of 30 spendable points upon full completion.

Q8: How is sequential upgrade order enforced?

Answer: `processUpgradePurchase()` calculates `currentStage = calculateRestorationStage(purchasedUpgradeIds)`. It then retrieves `nextUpgrade = CAMPUS_UPGRADES[currentStage]`. A student cannot purchase Stage 2 (Solar) before Stage 1 (Trees) because the system only evaluates affordability and eligibility for the exact next sequential upgrade in the array.

Q9: Explain the mathematical logic behind level progression in `calculateLevelInfo`.

Answer:

- **Level 1 (0–49 pts):** Progress is `(pts / 50) * 100%`.
- **Level 2 (50–99 pts):** Progress is `((pts - 50) / 50) * 100%`.
- **Level 3 (100–169 pts):** Span is 70 points. Progress is `((pts - 100) / 70) * 100%`.
- **Level 4 (170–210 pts):** Span is 40 points. Progress is `((pts - 170) / 40) * 100%`. At 210, `isCompletedProgression` becomes `true`.

Q10: Why is the Earth Guardian badge threshold (160 pts) different from the Earth Guardian level name (100–169 pts)?

Answer:

- In `LEVELS`, **Level 3** is titled "Earth Guardian" and covers the broad range from 100 to 169 points.
- In `BADGES`, the **Earth Guardian Badge** (`bdg-160`) is an advanced milestone trophy awarded at 160 points to reward students nearing the end of Level 3 before they break into Level 4 (Planet Champion at 170 pts). They share a thematic name but represent distinct progression achievements.

Q11: How does the dynamic leaderboard sorting and tie-breaking work?

Answer: In `calculateLeaderboard(lifetimePoints)`, we clone the roster and set Aditya's score to the current `lifetimePoints`. We then sort using JavaScript's `.sort()`:

```
const sorted = participants.sort((a, b) => {
  if (b.pts !== a.pts) {
    return b.pts - a.pts; // Primary: descending score
  }
  return a.initialIndex - b.initialIndex; // Tie-breaker: original roster seniority
});
```

When Aditya reaches 210 points, he ties with Priya Nair (210 pts). Because Priya's `initialIndex` is 2 and Aditya's is 7, Priya deterministically keeps Rank #3 (Bronze podium), and Aditya takes Rank #4.

Q12: Explain the startup hydration lifecycle in `GameContext.js`.

Answer:

1. Component mounts with `isHydrated = false` and empty ID arrays.
2. `AppContent` detects `!isHydrated` and renders a clean loading card with an `ActivityIndicator`.
3. `useEffect` calls `storage.getItem('ecoquest.gameState.v1')`.
4. If found, the JSON string is parsed and passed to `validateSavedGameState()`.

5. Sanitized IDs update state and refs.
6. `finally` block sets `isHydrated = true`, and the main 5-tab interface renders seamlessly without UI flicker.

Q13: What happens if corrupt or malformed JSON is found in AsyncStorage?

Answer: `validateSavedGameState(raw)` wraps parsing in a `try...catch` block. If parsing fails, or if the stored data is not an object, it logs a warning and returns `{ valid: false, reason: 'malformed_json', completedQuestIds: [], purchasedUpgradeIds: [] }`. The app safely defaults to a clean baseline state rather than crashing.

Q14: How does the app handle unknown or tampered IDs in storage?

Answer: `validateSavedGameState()` validates every quest ID against `knownQuestIds = new Set(QUESTS.map(q => q.id))`. Any unrecognized or duplicate ID is discarded. For upgrades, it verifies that they match the strict sequential order of `CAMPUS_UPGRADES`. If an out-of-order ID is found, validation stops at that point, guaranteeing internal integrity.

Q15: What is the purpose of `saveChainRef`?

Answer: React state updates are asynchronous, and multiple rapid user interactions could trigger overlapping calls to `AsyncStorage.setItem`. `saveChainRef` maintains an ordered Promise chain (`saveChainRef.current = saveChainRef.current.then(...)`). This acts as an asynchronous mutex, ensuring writes execute sequentially and newer state never gets overwritten by an older pending write.

Q16: Why are `completedQuestIdsRef` and `purchasedUpgradeIdsRef` used alongside `useState`?

Answer: React state updates do not reflect immediately in the current execution frame—they are scheduled for the next render. By keeping an immediate, synchronous `useRef` updated in tandem, rapid sequential calls (e.g., clicking two quests in quick succession) can read the true immediate state from `ref.current` without race conditions.

Q17: What happens if an AsyncStorage write fails during runtime?

Answer: `queuePersist` wraps the storage call in a `try...catch` block. If storage fails (e.g., disk quota or system error), it catches the exception, logs the error to the console, and returns `false`. Crucially, it does not throw an unhandled exception, allowing the user's in-memory session to continue functioning smoothly.

Q18: Explain the complete sequence of `resetDemo()`.

Answer:

1. Checks that the app is hydrated and not already resetting (`isResetting` guard).
2. Sets `isResetting = true`.
3. Awaits any pending writes in `saveChainRef`.
4. Calls `storage.removeItem('ecoquest.gameState.v1')`.
5. Clears `completedQuestIdsRef.current = []` and `purchasedUpgradeIdsRef.current = []`.
6. Sets both React state arrays to empty `[]`.
7. Sets `isResetting = false` and returns `{ success: true }`.
8. UI instantly snaps back to Stage 0, Level 1, 0 points, locked badges, and Rank #8.

Q19: Why does EcoQuest not require an external backend or database server?

Answer: For the scope of a college Community Engagement demonstration, the core goal is demonstrating client-side gamification mechanics, incentive modeling, and mobile responsiveness. Running entirely offline

eliminates network latency, server hosting costs, API failure points, and connectivity requirements during college evaluations.

> **Q20: Why are the leaderboard peers simulated rather than live online students?**

Answer: Implementing a live multi-tenant leaderboard would require a cloud database (e.g., Firebase or PostgreSQL), authentication, network security, and active internet connectivity during the viva. Using deterministic, realistic campus peer profiles allows examiners to directly verify dynamic ranking algorithms and tie-breaking logic without relying on external Wi-Fi.

> **Q21: How are visual layers created in `CampusScene.js` without external image assets?**

Answer: `CampusScene.js` uses native React Native components with CSS-style properties: absolute positioning, z-indexing, border radii, linear backgrounds, and emoji symbols. This provides crisp, vector-like visuals at zero file weight with zero network loading delay.

> **Q22: What is the time complexity of the calculation functions?**

Answer:

- `calculateLifetimePoints`: $O(N)$ where $N \le 8$ quests.
- `calculateSpendablePoints`: $O(N + M)$ where $N \le 8$ quests and $M \le 4$ upgrades.
- `calculateLeaderboard`: $O(P \log P)$ where $P = 8$ participants.

Because data sets are small and bounded, every selector computes in sub-millisecond time ($< 0.1\text{ ms}$), providing 60 FPS smooth rendering.

Section E: Calculations the Examiner May Ask

Examiners often test your mastery of the numbers. Be prepared to show these exact step-by-step calculations.

Calculation 1: Sum of All Eight Quest Rewards

$$\text{Total Lifetime Points} = \sum_{i=1}^{8} \text{Reward}(\text{Quest}_i)$$

Quest ID	Title	Category	Points	Running Total
`ch-1`	Reusable Water Bottle	Waste	30	30
`ch-2`	Lights & Fans Off	Energy	20	50
`ch-3`	Green Commute	Transport	40	90
`ch-4`	Plant-Based Meal	Food	25	115
`ch-5`	Segregate Campus Waste	Waste	35	150
`ch-6`	5-Minute Shower	Water	25	175
`ch-7`	Unplug Idle Chargers	Energy	15	190
`ch-8`	Bring a Tote Bag	Waste	20	**210**

$$30 + 20 + 40 + 25 + 35 + 25 + 15 + 20 = 210 \text{ Lifetime Points}$$

Calculation 2: Total Cost of All Four Campus Upgrades

$$\text{Total Upgrade Cost} = \sum_{j=1}^{4} \text{Cost}(\text{Upgrade}_j)$$

Stage	Upgrade ID	Project Title	Cost	Running Cost
Stage 1	`upg-trees`	Native Shade Trees	30	30
Stage 2	`upg-solar`	Rooftop Solar Array	40	70
Stage 3	`upg-compost`	Smart Compost Hubs	50	120
Stage 4	`upg-water`	Pure Water Cistern	60	**180**

$$30 + 40 + 50 + 60 = 180 \text{ Points Total Cost}$$

Calculation 3: Final Spendable Surplus

$$\text{Spendable Points} = \text{Lifetime Points} - \text{Total Upgrade Cost}$$

$$\text{Surplus} = 210 - 180 = 30 \text{ Spendable Points Surplus}$$

Explanation: After funding every single restoration project on campus, the student is not left at zero; they retain a 30-point surplus in their account.

Calculation 4: Specific Mid-Game Economy Example

Examiner asks: "Suppose a student completes `ch-1` (30 pts) and `ch-4` (25 pts), then purchases Stage 1 Native Shade Trees (30 pts). What are their lifetime and spendable points?"

1. Quests Completed:

$$\text{Lifetime Points} = 30\,(\text{ch-1}) + 25\,(\text{ch-4}) = \mathbf{55\text{ Lifetime Points}}$$

2. Upgrade Purchased:

$$\text{Total Upgrade Cost} = 30\,(\text{upg-trees})$$

3. Spendable Points:

$$\text{Spendable} = \text{Lifetime} - \text{Cost} = 55 - 30 = \mathbf{25\text{ Spendable Points}}$$

4. Current Status:

* Lifetime Points: **55 pts** (Player is Level 2: Green Scout).

* Spendable Points: **25 pts**.

* Campus Stage: **Stage 1** (Trees planted).

* Next Upgrade: Rooftop Solar Array (Cost: 40 pts).

* Shortfall to Next Upgrade: $40 - 25 = \mathbf{15\text{ points needed}}$.

Calculation 5: Player Level Threshold Transitions

Level Tier	Name	Point Range	Next Threshold	Span	Progress Formula
Level 1	Eco Seedling	0 – 49 pts	50	50 pts	`(pts / 50) * 100%`
Level 2	Green Scout	50 – 99 pts	100	50 pts	`((pts - 50) / 50) * 100%`
Level 3	Earth Guardian	100 – 169 pts	170	70 pts	`((pts - 100) / 70) * 100%`
Level 4	Planet Champion	170 – 210 pts	210 (Max)	40 pts	`((pts - 170) / 40) * 100%`

Exact Transition Points:

- At **49 points:** Level 1 Eco Seedling (Progress: 98%).
- At **50 points: Level 2 Green Scout** (Progress: 0%).
- At **99 points:** Level 2 Green Scout (Progress: 98%).
- At **100 points: Level 3 Earth Guardian** (Progress: 0%).
- At **169 points:** Level 3 Earth Guardian (Progress: 99%).
- At **170 points: Level 4 Planet Champion** (Progress: 0%).
- At **210 points:** Level 4 Planet Champion (Progress: 100%, `isCompletedProgression = true`).

Calculation 6: Earth Guardian Badge vs. Level 4 Threshold

Examiner asks: "Why do you have 160 points for Earth Guardian and 170 points for Level 4?"

- **Badge `bdg-160` (Earth Guardian Trophy):** Unlocks automatically when Lifetime Points ≥ 160.

* At 160 points, the student is still in **Level 3** (span 100–169).

* This milestone badge rewards deep commitment right before reaching top-tier status.

- **Level 4 Transition (Planet Champion):** Requires ≥ 170 points.

* At 160 points, student needs $170 - 160 = \mathbf{10\text{ more points}}$ to reach Level 4.

* Thus, badge unlock and level promotion are intentionally decoupled milestones.

Calculation 7: Dynamic Leaderboard Climbing & Tie-Breaking

Baseline Roster Points:

1. Maya Sharma: 340 pts 🥇
2. Aarav Patel: 290 pts 🥈
3. Priya Nair: 210 pts 🥉
4. Rohan Das: 160 pts
5. Zara Khan: 120 pts
6. Vikram Rao: 75 pts
7. Neha Gupta: 40 pts
8. **Aditya (User): 0 pts (Rank #8)**

Simulation Sequence:

- **At 0 pts:** User is **Rank #8** ($0 < 40$).
- **At 30 pts (ch-1):** User is **Rank #8** ($30 < 40$).
- **At 50 pts (ch-1 + ch-2):** User passes Neha (40 pts) $\rightarrow$ climbs to **Rank #7**.
- **At 90 pts (+ ch-3):** User passes Vikram (75 pts) $\rightarrow$ climbs to **Rank #6**.
- **At 150 pts (+ ch-4 + ch-5):** User passes Zara (120 pts) $\rightarrow$ climbs to **Rank #5**.
- **At 175 pts (+ ch-6):** User passes Rohan (160 pts) $\rightarrow$ climbs to **Rank #4**.
- **At 210 pts (all 8 quests completed):** User reaches 210 points, equaling Priya Nair (210 pts).

* Tie-Breaker Rule: `a.initialIndex - b.initialIndex`.

* Priya Nair was in the initial roster at index 2 (`p-3`).

* Aditya was registered at index 7 (`user`).

* Priya Nair deterministically keeps **Rank #3 (Bronze Medal 🥉)**.

* Aditya takes **Rank #4**.

Calculation 8: Invariant Proof — Why Upgrades Cannot Reduce Lifetime Points

$$\text{Lifetime Points} = \sum_{q \in \text{completedQuestIds}} \text{Reward}(q)$$

$$\text{Spendable Points} = \max(0, \text{Lifetime Points} - \sum_{u \in \text{purchasedUpgradeIds}} \text{Cost}(u))$$

- In JavaScript, `calculateLifetimePoints` only takes `completedQuestIds` as an argument.
- Purchasing an upgrade invokes `purchasedUpgradeIds.push(nextUpgrade.id)`.
- It does not mutate `completedQuestIds`.
- Therefore, mathematically and programmatically, purchasing upgrades has **zero effect on lifetime points**.

Calculation 9: Invariant Proof — Why Spending Points Does Not Demote Level or Rank

- `calculateLevelInfo(lifetimePoints)` only receives `lifetimePoints`.
- `calculateLeaderboard(lifetimePoints)` only receives `lifetimePoints`.
- Neither function receives `spendablePoints` or `purchasedUpgradeIds`.
- Therefore, a student with 180 points who spends all 180 points on campus upgrades remains Level 4 and retains Rank #4 on the leaderboard. Spending points funds physical restoration without penalizing personal academic standing.

Section F: Difficult or Critical Examiner Questions

These are realistic, challenging questions examiners ask to test your depth of understanding and honesty.

> **1. Does your application prove that students actually performed these green actions in the real world?**

Defense: "No, Sir/Madam. In this prototype, quest completion is self-reported through in-app user action. The purpose of this project is to model the gamified incentive structure and UI/UX feedback loops. Validating physical behavior would require external hardware—such as AI image verification, smart meter telemetry, or barcode scanners at waste bins—which was outside the academic scope of this client demonstration."

> **2. Does the app measure genuine carbon emission reductions or liters of water saved?**

Defense: "No. The environmental impact labels in the quest cards (e.g., 'Medium Impact', 'Conserves classroom electricity') are educational estimates designed for student awareness. They are pedagogical benchmarks, not laboratory-certified scientific measurements."

> **3. Why are the other student scores on the leaderboard fictional?**

Defense: "The peer scores represent fixed benchmark profiles designed to demonstrate real-time ranking algorithms, podium assignment, and deterministic tie-breaking. Building a live multi-user network would require cloud servers, user authentication, and persistent network access, introducing external failure points during a live viva examination."

> **4. Why did you not implement a cloud backend using Firebase, Node.js, or Supabase?**

Defense: "We intentionally architected EcoQuest as an offline-first mobile application. This decision guarantees zero network latency, zero hosting costs, and total reliability during college presentations, where campus Wi-Fi can be unstable. For a single-user prototype demonstrating client-side gamification, local persistence using AsyncStorage was the most robust and appropriate engineering choice."

> **5. What happens if the student clears the app storage or uninstalls the app?**

Defense: "Because progress is stored locally in AsyncStorage on the device, clearing app data or uninstalling the app will erase the saved JSON file (`ecoquest.gameState.v1`). The app will cleanly initialize a fresh demo session starting at Stage 0 and 0 points. In a production release, this would be addressed by backing up user state to a cloud database."

> **6. What prevents a student from falsely claiming all eight quests in 10 seconds?**

Defense: "In the current demonstration prototype, nothing prevents rapid button tapping because quests are self-reported. However, in our architecture, each quest can only be completed once—the engine strictly prevents duplicate point awards. In a future production release, we would add daily time-locks, GPS geofencing around campus recycling stations, and photo verification."

> **7. What prevents race conditions if a student taps buttons very quickly?**

Defense: "We implemented two safeguards in `GameContext.js` :

First, synchronous React `useRef` instances (`completedQuestIdsRef` and `purchasedUpgradeIdsRef`) hold the immediate state and are checked before any action proceeds.

Second, all storage writes are queued through `saveChainRef` , an asynchronous Promise chain that serializes disk writes one after another, preventing older asynchronous writes from overwriting newer state."

8. What happens if the device storage is full and AsyncStorage fails?

Defense: "All storage calls are wrapped in `try...catch` blocks. If an AsyncStorage write fails, the error is caught and logged to the console without crashing the mobile app. The user's in-memory session continues uninterrupted, preserving app stability."

9. Why did you separate Lifetime Points from Spendable Points?

Defense: "In behavioral economics and game design, if spending currency demotes a user's status—for example, if buying a campus tree causes a student to drop from Level 3 to Level 1 and fall down the leaderboard—users will hoard points and refuse to participate in campus restoration. Separating lifetime achievement from spendable currency solves this dilemma."

10. How do you know your calculations work correctly without human error?

Defense: "We implemented an automated integration test suite in `tests/test_integration.mjs` containing 83 comprehensive assertions. It programmatically tests every quest reward, upgrade cost, level threshold, badge unlock rule, storage serialization, and demo reset scenario. All 83 tests pass with 100% success rate."

11. What is the difference between automated tests and physical-device testing?

Defense: "Automated tests verify pure logic, state transitions, mathematical balance, and serialization in a headless Node.js environment. Physical-device testing in Expo Go verifies user-facing behavior: touchscreen responsiveness, layout rendering across different screen aspect ratios, font scaling, and safe-area clearance over Android system navigation bars."

12. What are the key limitations of relying solely on AsyncStorage?

Defense: "AsyncStorage is unencrypted, has storage size quotas (typically 6 MB on Android), is tied strictly to the local physical device, and does not support relational SQL queries. While perfect for storing our lightweight JSON configuration (~500 bytes), an enterprise release would require SQLite or a remote cloud database."

13. How could this project scale to a real college deployment of 5,000 students?

Defense: "To scale to a live college deployment, we would:

1. Replace AsyncStorage with a secure backend API (e.g., Node.js / Express with PostgreSQL).
2. Implement college student email authentication (SSO / OAuth).
3. Introduce daily recurring quest cycles with cooldown timers.
4. Add image-verification using AI vision models for quest validation.
5. Create an administrative dashboard for campus faculty to monitor collective sustainability metrics."

14. What would be the very first architectural change you would make to support multiple real users?

Defense: "The very first change would be introducing a backend authentication service. In `GameContext.js`, we would replace the local `storage.getItem` and `storage.setItem` calls with authenticated REST or GraphQL API calls (`fetch('/api/user/progress')`), passing a JWT bearer token to store progress securely against a unique student database record."

15. Why did you use React Native rather than Flutter or native Android (Kotlin)?

Answer: "React Native leverages the JavaScript and React ecosystem, which enables code reusability, a vast open-source library collection, and rapid declarative UI prototyping. For a student computer science project, it allowed us to build a high-performance native Android application while focusing our efforts on game architecture, state derivation, and user experience."

> **16. What was the most difficult bug you encountered during development, and how did you resolve it?**

Defense: "The most challenging issue was a race condition during rapid quest completion and demo resets, where asynchronous `AsyncStorage.setItem` calls would occasionally interleave, causing older state to overwrite newer state. We solved this by implementing `saveChainRef` —a chained Promise queue that acts as an asynchronous write mutex—and backing React state with synchronous `useRef` trackers to guarantee immediate atomic consistency."

Section G: Module-by-Module Code Walkthrough

Be prepared to explain the exact purpose and mechanics of every source file.

```
ecoquest-demo/
├── App.js                  # Root Component & Hydration Guard
├── src/
│   ├── data/
│   │   └── gameData.js     # Single Source of Truth for Data Definitions
│   ├── context/
│   │   └── GameContext.js  # Central State, Economy Engine & Persistence
│   ├── components/
│   │   ├── BottomNavBar.js  # 5-Tab Navigation with Android Insets
│   │   └── CampusScene.js   # Native Illustrated Quad Environment (Stages 0-4)
│   ├── screens/
│   │   ├── HomeScreen.js    # Campus World, Level Progress & Upgrades
│   │   ├── QuestsScreen.js  # 8 Sustainability Quests & Category Filtering
│   │   ├── RankScreen.js    # 8-Participant Leaderboard & Podium
│   │   ├── BadgesScreen.js  # 8 Achievement Trophies & Unlock Progress
│   │   └── ProfileScreen.js # Student Identity Card & Confirmed Demo Reset
│   └── theme/
│       └── colors.js        # Emerald and Slate Design System Palette
```

1. `App.js`

- **Primary Responsibility:** Root application entry point. Configures safe area providers, renders the loading splash screen, and manages the active tab view.
- **Why Separate:** Keeps root initialization and top-level view orchestration separate from game state logic.
- **Key Inputs & Outputs:** Renders `SafeAreaProvider` and `GameProvider`. Uses `useGame()` to read `isHydrated`.
- **Interaction:** If `!isHydrated`, it renders a themed splash card with `ActivityIndicator`. Once hydrated, it renders the active screen component and passes `activeTab` to `BottomNavBar`.

2. `src/context/GameContext.js`

- **Primary Responsibility:** Central state manager, derivation engine, persistence coordinator, and reset controller.
- **Why Separate:** Encapsulates all game logic in one place so individual UI screens remain pure presentation components.
- **Key Exports:** `GameProvider`, `useGame()`, `STORAGE_KEY`, `SCHEMA_VERSION`, `validateSavedGameState()`, `serializeGameState()`, `calculateLifetimePoints()`, `calculateSpendablePoints()`, `calculateRestorationStage()`, `calculateLevelInfo()`, `calculateBadges()`, `calculateLeaderboard()`, `processQuestCompletion()`, `processUpgradePurchase()`.
- **Interaction:** Wraps the app. Exposes all reactive state and handlers (`completeQuest`, `purchaseNextUpgrade`, `resetDemo`) to consumer screens via the `useGame()` hook.

3. `src/data/gameData.js`

- **Primary Responsibility:** Central repository of static configuration arrays.
- **Why Separate:** Prevents hardcoding quest names, point values, upgrade costs, or badge criteria across multiple screen files.
- **Key Exports:** `QUEST_CATEGORIES`, `QUESTS` (8 items), `CAMPUS_UPGRADES` (4 items), `LEVELS` (4 tiers), `BADGES` (8 items), `INITIAL_LEADERBOARD` (8 participants).
- **Interaction:** Imported by `GameContext.js` for calculation selectors, and imported by screens for labels, icons, and categories.

4. `src/components/CampusScene.js`

- **Primary Responsibility:** Renders the illustrated campus quad using pure React Native layout primitives.
- **Why Separate:** Isolates the complex visual rendering of trees, lawns, solar panels, compost pods, and water fountains from business logic.
- **Key Props:** `restorationStage` (number 0 to 4).
- **Interaction:** Receives `restorationStage` from `HomeScreen.js`. As stage increases, it conditionally renders visual elements and updates sky and lawn colors using standard CSS-in-JS style layering.

5. `src/components/BottomNavBar.js`

- **Primary Responsibility:** Renders the persistent bottom tab navigation bar.
- **Why Separate:** Reusable navigation widget cleanly decoupled from screen content.
- **Key Props:** `activeTab` (string), `onSelectTab` (callback function).
- **Interaction:** Calls `useSafeAreaInsets()` to calculate dynamic bottom padding on Android devices, ensuring buttons clear 3-button navigation and gesture pills. Triggers `onSelectTab(tabId)` when a tab is tapped.

6. `src/screens/HomeScreen.js`

- **Primary Responsibility:** Central dashboard. Displays the `CampusScene`, student level card, points overview, and next restoration upgrade card.
- **Why Separate:** Dedicated screen for home view presentation.
- **Key State Consumed:** `lifetimePoints`, `spendablePoints`, `restorationStage`, `nextUpgrade`, `canAffordNextUpgrade`, `levelInfo`, `userRank`.
- **Interaction:** Dispatches `purchaseNextUpgrade()` when the student funds a project. Navigates to Quests via the quick-action button.

7. `src/screens/QuestsScreen.js`

- **Primary Responsibility:** Displays the 8 sustainability quests with category filtering.
- **Why Separate:** Dedicated screen for micro-action discovery and quest execution.
- **Key State Consumed:** `quests`, `completedQuestIds`, `completeQuest()`.
- **Interaction:** Local state `selectedCategory` filters quest cards. Tapping 'Complete Quest' triggers `completeQuest(quest.id)`, instantly locking the button and awarding points.

8. `src/screens/RankScreen.js`

- **Primary Responsibility:** Displays the 8-participant campus leaderboard and Top-3 podium.
- **Why Separate:** Dedicated screen for social benchmarking and competition visualization.
- **Key State Consumed:** `leaderboard`, `userRank`, `top3Podium`, `lifetimePoints`.
- **Interaction:** Automatically reflects real-time rank updates as the demonstrator earns points, highlighting the user's row and updating podium cards.

9. `src/screens/BadgesScreen.js`

- **Primary Responsibility:** Displays the 8 achievement trophy cards and completion progress bar.
- **Why Separate:** Dedicated screen for long-term milestone tracking.
- **Key State Consumed:** `badges`, `unlockedBadgeCount`.
- **Interaction:** Reads evaluated badge objects from `GameContext.js`, displaying unlocked cards in emerald and locked cards in subdued gray.

10. `src/screens/ProfileScreen.js`

- **Primary Responsibility:** Displays student profile metadata, cumulative statistics, and the confirmed demo reset workflow.
- **Why Separate:** Dedicated screen for account inspection and presentation maintenance.
- **Key State Consumed:** `lifetimePoints`, `completedQuestCount`, `levelInfo`, `unlockedBadgeCount`, `resetDemo()`, `isResetting`, `isHydrated`.
- **Interaction:** Tapping 'Reset Demo' triggers a native `Alert.alert` dialog. Confirming executes `resetDemo()`, cleanly clearing storage and state.

Section H: Live Demonstration Script & Fallback Plan

A rehearsable 5–7 minute walkthrough script with exact screen actions and spoken talking points.

Step 1: Introduction & Initial Baseline (1 Minute)

- **Action:** Hold the phone showing the **Home Tab**.
- **Spoken Script:**

> "Examiners, here is EcoQuest running live on an Android device via Expo Go. Notice our starting state: the campus quad is at **Stage 0 (Dormant Quad)** with dry beige grounds. My account starts at **Level 1: Eco Seedling** with **0 Lifetime Points** and **0 Spendable Points**, and our initial campus standing is **Rank #8**."

Step 2: Complete First Quest in Quests Tab (1 Minute)

- **Action:** Tap the **Quests Tab** (🎯). Point to the categories. Filter by 'Waste'. Tap **'Complete Quest (+30 pts)'** on *Reusable Water Bottle*.
- **Spoken Script:**

> "Let's open the Quests tab. We have eight challenges across Energy, Waste, Water, Transport, and Food. I will complete our first waste quest: 'Reusable Water Bottle'.
>
> When I tap 'Complete Quest', notice the instant feedback: the card locks with a green checkmark, preventing duplicate rewards. Our points summary immediately shows **30 Lifetime Points** and **30 Spendable Points**."

Step 3: Verify Trophy Badge Unlock (30 Seconds)

- **Action:** Tap the **Badges Tab** (🎖️).
- **Spoken Script:**

> "Navigating to the Badges tab, our first trophy—**Eco Starter (🌱)**—has automatically unlocked because its condition of 'Complete 1 Quest' is satisfied. The other 7 badges remain locked with clear progress counters."

Step 4: Fund Stage 1 Campus Upgrade (1 Minute)

- **Action:** Return to the **Home Tab** (🏠). Show that the 'Fund Project (30 pts)' button is now vibrant emerald. Tap **'Fund Native Shade Trees (30 pts)'**.
- **Spoken Script:**

> "Returning to the Home tab, our restoration card now shows that **Native Shade Trees** is affordable. It costs 30 points.
>
> When I tap 'Fund Upgrade', watch the campus illustration: the ground transforms from dry beige to lush green, and native oak trees appear in the quad.
>
> Now observe our economic balance: **Lifetime Points remain 30**, but **Spendable Points drop to 0**. Our academic standing was not penalized for restoring the campus."

Step 5: Advance Progression & Demonstrate Leaderboard Climb (1 Minute)

- **Action:** Switch to **Quests Tab**. Complete *Green Commute* (Transport, +40 pts). Lifetime points reach 70.
- **Action:** Switch to **Rank Tab** (🏆).
- **Spoken Script:**

> "Let's complete another quest: 'Green Commute' for 40 points. Our lifetime points reach **70 points**.

>

> Notice two immediate updates:

> First, our player level promoted from Eco Seedling to **Level 2: Green Scout** (threshold was 50 points).

> Second, on the Campus Leaderboard, my account has climbed from **Rank #8 to Rank #6**, surpassing both Neha Gupta (40 pts) and Vikram Rao (75 pts). The leaderboard recalculates and resorts in real time."

Step 6: Demonstrate Offline Persistence Survival (1 Minute)

- **Action:** Press the Android Home / Overview button. Swipe away / close the EcoQuest app, or press `r` in the Metro terminal to reload. Reopen EcoQuest.
- **Spoken Script:**

> "Now I will demonstrate persistence. I am closing and reloading the application.

>

> Notice the brief loading splash card: during this moment, `GameContext` reads our serialized JSON from `AsyncStorage`. As it resolves, our progress is completely preserved: **Stage 1 campus remains restored**, **70 points remain intact**, and our completed quests stay locked. Progress survives app restarts without needing any internet connection."

Step 7: Demonstrate Two-Step Confirmed Demo Reset (1 Minute)

- **Action:** Navigate to the **Profile Tab** (👤). Scroll to the bottom. Tap **'Reset Demo Data'**. When the alert appears, tap **'Cancel'** to show safety. Tap **'Reset Demo Data'** again, then tap **'Reset Everything'**.
- **Spoken Script:**

> "Finally, for live demonstrations, we need a way to reset state for the next evaluator. In the Profile tab, tapping 'Reset Demo Data' opens a native safety confirmation dialog. Tapping 'Cancel' keeps our data safe.

>

> When I tap 'Reset Everything', the AsyncStorage key is removed, state arrays are cleared, and the app cleanly returns to its pristine baseline: **Stage 0 dormant campus, 0 points, Level 1, and Rank #8**."

Step 8: Demonstration Fallback Plan

Be prepared for technical glitches with this contingency protocol:

Scenario	Immediate Action	Spoken Explanation
Expo Go disconnects from Wi-Fi	Turn on Mobile Hotspot on phone, connect laptop to phone hotspot, restart Expo with `npx expo start`.	"Examiners, campus Wi-Fi dropped. I am switching to my phone's local hotspot, which provides an isolated direct local network connection."
Metro bundler cache gets stuck	Run `npx expo start --go -c` in terminal to clear bundler cache.	"I am clearing the Metro bundler cache using the `-c` flag to force a clean bundle rebuild."
Phone battery dies or screen fails	Run `npx expo start --web` or open Chrome to `http://localhost:8081` on laptop.	"I will demonstrate the identical React Native component tree in the browser while my device charges."
Complete hardware failure	Open the pre-compiled `submission/EcoQuest_Project_Report.pdf` and show high-resolution verified UI screenshots and integration test proofs.	"I will walk you through the architectural screenshots and verified automated test suite in our submission report."

Section I: 30-Second Closing Statement

Deliver this strong, confident conclusion at the end of your viva.

"To conclude, Respected Examiners:

EcoQuest demonstrates how modern computer science—specifically mobile application development, reactive state management, and gamification mechanics—can transform passive sustainability guidelines into measurable daily student action.

In this project, I engineered a robust two-tier economy, a dynamic campus illustration, offline persistence, and automated test verification with 83 passing assertions.

While automated IoT sensor verification and cloud deployment remain future enhancements, EcoQuest successfully delivers a polished, responsive, and mathematically balanced mobile prototype for campus environmental stewardship.

Thank you, and I am happy to answer any further questions."